# Jack T. Dinsmore

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## Research Interests

- Winds, outflows, and nebulae surrounding compact objects
- Cosmic rays and magnetic fields in the interstellar medium
- Advancing analysis techniques for X-ray and optical astronomical data, including polarization

## Education

Stanford University — 2022 – 2027 (projected)
Ph.D. in Physics (GPA: 4.0/4.0)
Advisor: Prof. Roger W. Romani

Massachusetts Institute of Technology — 2018 – 2022
B.S. in Physics, Minors in Astronomy and Mathematics, Concentration in Music (GPA: 5.0/5.0)
Research Mentors: Profs. Tracy R. Slatyer, Julien de Wit, and Philip Harris

## Awards & Honors

- NSF Graduate Research Fellowship Program Honorable Mention — 2024
- Inducted into ΦBK and ΣΠΣ honors societies — 2022
- MIT Barrett Prize for excellence in astrophysics research — 2022
- Lehigh University REU program — 2020

## Telescope Allocations

Observing record: 11 successful grant proposals (10 as co-I, 1 as PI). $>$2 Ms X-ray observatory time, $\sim$8 nights optical observing experience.

- *NuSTAR*: 500 kiloseconds (1 grant as co-I)
- Imaging X-ray Polarimetry Explorer: 1 megasecond (1 grant as co-I)
- *Chandra* X-ray observatory, ACIS: 510 kiloseconds (3 grants as co-I)
- Magellan Observatory, proto-Lightspeed: 6 nights (3 grants as co-I)
- National Radio Astronomy Observatory: 50 minutes (1 grant as co-I)

## Additional Grant Allocations

- Kavli Institute: KICK grant for developing new technology (1 grant as PI)
- KIPAC: Innovation grant (1 grant as co-I)

## Research Mentoring

- Stanford undergraduate student Calvin McCullumsmith — 2026
- Stanford undergraduate student Tristan Jau-Shi Lu — 2026
- Stanford graduate student Ningyue (Fanny) Fan — 2026
- Stanford undergraduate student Peter Lindholm — 2025
- Stanford graduate student Kaitlyn Karpovich — 2024

## Teaching

- Mentor, Stanford directed reading program 2025
- TA, Stanford PHYSICS 130: Quantum Mechanics (rated 4.41/5 in effectiveness) Spring 2025
- TA, Stanford PHYSICS 110: Advanced Mechanics (rated 4.36/5 in effectiveness) Fall 2024
- TA, Stanford PHYSICS 120: Intermediate E&M (rated 4.73/5 in effectiveness) Winter 2024
- TA, Stanford PHYSICS 44: Lab for E&M (rated 4.71/5 in effectiveness) Spring 2023
- Course material designer and TA, MIT 8.S50: Statistics Winter 2022
- Grader, MIT 8.012: Physics I Spring 2019
- Teacher, MIT Academic Teaching Initiative 2018

## Professional Service & Outreach

- Reviewer for the Astrophysical Journal 2026
- Member, KIPAC Tea committee 2025–present
- Member, KIPAC communications director search committee 2026
- Editor, Writer, and Web Manager for the KIPAC Research Highlights 2023 – present
- Co-organizer, KIPAC computing bootcamp 2022
- Volunteer, KIPAC stargazing 2025
- Volunteer, KIPAC community day (science fair) 2025
- Volunteer, Stempalooza (science fair) 2024
- Mentor, Stanford Future Advancers of Science and Technology 2023 – 2024
- Organizer, KIPAC community day (science fair) 2023

## Press Coverage

- NASA press release "NASA Space Telescope Maps Magnetic Fields of 'Lighthouse' Pulsar" 2026
- "KIPAC Year in Review" 2024—2025
- "KIPAC Years in Review" 2022—2024
- MIT News: "Method for decoding asteroid interiors could help aim asteroid-deflecting missions" 2022
- Astrobites: "Using tides to peak into asteroid interiors" 2022

## Invited Presentations

- "IXPE Polarizations of the Lighthouse X-ray Filament," MAT talk (MIT) 2026
- "The Lighthouse Pulsar and its Polarized X-ray Filament," Tea talk (Caltech) 2026
- "proto-Lightspeed: an MKI-KIPAC Project to Capture the Changing Polarized Sky," Kavli Symposium (University of Cambridge) 2026
- "proto-Lightspeed: a Low Noise, High Speed Imager for Magellan," KIPAC Tea (Stanford) 2026
- "IXPE Polarimetry of Pulsar Wind Nebulae," Friends of IXPE meeting (remote) 2026

## Contributed Presentations

- American Astronomical Society Summer meeting (Pasadena, CA) 2026
- IXPE PWN group (remote) 2026
- Compact Objects Group, (Stanford) 2026
- Massachusetts Institute of Technology CMOS group (MIT) 2025
- Imaging X-ray Polarimetry Explorer SASWG group (remote) 2025
- International X-ray Polarimetry Symposium (Huntsville, AL) 2024
- Fields, Flows, and Filaments in the Magnetic ISM (Stanford) 2024
- Asteroids, Comets, and Meteorites conference (Flagstaff, AZ) 2023
- Imaging X-ray Polarimetry Explorer SASWG group (remote) 2023
- Apophis T−7 Years (remote) 2022

## Lead-author Refereed Publications

Publication record: 19 total peer-reviewed articles, two published software libraries. Cited in ∼210 academic works. $h$-index of 6.

9. **Jack T. Dinsmore**. A Statistical Model for Stable Rings Beyond the Roche Limit. *Icarus (in press)*, 2026
8. **Jack T. Dinsmore**, Roger W. Romani, S. Zhang, C.-Y. Ng, Stefano Silvestri, and others. IXPE Polarizations of the Lighthouse Pulsar, Trail, and Filament. *ApJ*, 1005(2):237, July 2026
7. **Jack T. Dinsmore** and Roger W. Romani. Chandra Proper Motions and Milliarcsecond Astrometry of Nineteen Pulsars. *ApJ*, 1000(2):183, March 2026
6. **Jack T. Dinsmore** and Roger W. Romani. Advanced Weights for IXPE Polarization Analysis. *ApJ*, 993(2):173, November 2025
5. **Jack T. Dinsmore**, Roger W. Romani, Nikos Mandarakas, Dmitry Blinov, and Ioannis Liodakis. The Guitar Filament's Magnetic Field Revealed by Starlight Polarization. *ApJ*, 980(2):229, February 2025
4. **Jack T. Dinsmore** and Roger W. Romani. A Catalog of Pulsar X-Ray Filaments. *ApJ*, 976(1):4, November 2024
3. **Jack T. Dinsmore** and Roger W. Romani. Polarization Leakage and the IXPE Point-spread Function. *ApJ*, 962(2):183, February 2024
2. **Jack T. Dinsmore** and Julien de Wit. Constraining the Interiors of Asteroids Through Close Encounters. *MNRAS*, 520(3):3459–3475, 10 2022
1. **Jack T. Dinsmore** and Tracy R. Slatyer. Luminosity Functions Consistent with a Pulsar-Dominated Galactic Center Excess. *JCAP*, 06(06):025, 2022

## Lead Author Non-Refereed Publications and Software

3. **Jack T. Dinsmore** and Roger W. Romani. A Physical Model of Pulsar X-ray Filaments. *arXiv e-prints*, page arXiv:2603.20532, March 2026
2. **Jack T. Dinsmore**. jtdinsmore/leakagelib: Advanced Weights for IXPE Polarimetry., September 2025
1. **Jack T. Dinsmore**. jtdinsmore/ixpenn: IXPE Neural Net Event Reconstruction Pipeline., September 2025

## Contributing Author Refereed Publications

10. Josephine Wong, **Jack T. Dinsmore**, Roger W. Romani, Stefano Silvestri, Shumeng Zhang, and others. Kes 75 with IXPE: Detection of Nebular X-Ray Polarization and Change in Pulsar Lightcurve. *ApJ*, 1006(2):113, August 2026
9. Christopher Layden, Kevin Burdge, Gabor Furesz, Juliana García-Mejía, **Jack T. Dinsmore**, and others. proto-Lightspeed: A High-speed, Ultra-low Read Noise Imager on the Magellan Clay Telescope. *PASP*, 138(6):065001, June 2026
8. Paolo Soffitta, Niccolò Bucciantini, Josephine Wong, Denis González-Caniulef, Matteo Bachetti, and others. IXPE View of the Crab Pulsar Following the 2025 July 17 and August 6 Glitches. *ApJ*, 1001(2):219, April 2026
7. Emma T. Chickles, Joheen Chakraborty, Kevin B. Burdge, Vik S. Dhillon, Paul Draghis, and others. An eclipsing 8.56 minute orbital period mass-transferring binary. *ApJ*, 1000(2):237, March 2026
6. Andrew G. Sullivan, **Jack T. Dinsmore**, and Roger W. Romani. X-Ray Polarization of the Intrabinary Shock in Redback Pulsar J1723−2837. *ApJ*, 999(1):128, March 2026
5. Niccolò Di Lalla, Nicola Omodei, Niccolò Bucciantini, **Jack T. Dinsmore**, Nicolò Cibrario, and others. X-Ray Polarization Detection of the Pulsar Wind Nebula in G21.5–0.9 with IXPE. *ApJ*, 991(1):67, September 2025
4. Tobin M. Wainer, Gail Zasowski, Joshua Pepper, Tom Wagg, Christina L. Hedges, and others. Catalog of Integrated-light Star Cluster Light Curves in TESS. *AJ*, 166(3):106, August 2023
3. Josephine Wong, Roger W. Romani, and **Jack T. Dinsmore**. Improved Measurements of the IXPE Crab Polarization. *ApJ*, 953(1):28, July 2023
2. Jeffrey Krupa, Kelvin Lin, Maria Acosta Flechas, **Jack Dinsmore**, Javier Duarte, and others. GPU Coprocessors as a Service for Deep Learning Inference in High Energy Physics. *Mach. Learn.: Sci. Technol.*, 2(3):035005, April 2021
1. **Jack Dinsmore**, Patrick Draper, David Kastor, Yue Qiu, and Jennie Traschen. Schottky Anomaly of deSitter Black Holes. *Class. Quant. Grav.*, 37(5):054001, 2020

## Contributing Author Non-Refereed Publications

1. Christopher Layden, Kevin Burdge, John J. Piotrowski, Gábor Furesz, Juliana García-Mejía, and others. The Lightspeed project: high-speed, ultra-low read noise imaging and polarimetry for the Magellan Telescopes. In *Ground-based and Airborne Instrumentation for Astronomy XI*, volume 14149, page 141492I. International Society for Optics and Photonics, SPIE, 2026

References are available upon request. Updated August 24, 2026